

Priedas 2 – FNA 16 straipsnio vertinimai

Distribution Network Flexibility Need Indicators Pursuant to Article 16(4) of the FNA

Based on the flexibility needs assessment methodology and ESO load modeling data, the distribution network flexibility need indicators are presented below.

Location of Flexibility Need (Geographical and Voltage Level):

- Geographical location: Needs are modeled across the entire territory of Lithuania, linking them to specific physical locations through a topological tracking algorithm.
- Voltage level: Calculations are performed and aggregated at the 110 kV voltage level, focusing on all 200 specific 110 kV transformer substation nodes.

Direction of Flexibility Need:

- Upwards and downwards flexibility need calculations have been performed.
- Upward needs are calculated by assessing the capacity of 110 kV transformer substations under N-1 contingency mode.

Timeframe and Energy Volumes:

- Time granularity: Although the mathematical model calculates needs for each hour of the year (a sequence of 8,760 hours), the indicators are aggregated into annual time blocks for the target years.
- Total energy (Total MWh): This indicator is calculated as the sum of all hourly needs throughout the year.
- Economic criteria: Economic criteria were not evaluated.

Local Maximum Power (Max MW):

This is the highest hourly flexibility value (MW) recorded throughout the year at a specific node. These calculations were performed for each 110 kV transformer substation, and the report presents the sum of the maximum recorded values across all nodes.

Sistemos lankstumo poreikių rodikliai pagal FNA 16 straipsnį

lentelė 1 ERAA upward ramping needs detalization

	Scenario	2028	2030	2033	2035
Number of MTUs showing an uncovered need, h	„Be jungčių ir P2X“	151	123	115	102
	„Be jungčių“	121	95	47	45
Interval between uncovered needs, h	„Be jungčių ir P2X“	81	101,3	105,3	111,6
	„Be jungčių“	97,8	135,2	235,4	260,8
Volume of uncovered needs in terms of energy, GWh	„Be jungčių ir P2X“	18,87	14,68	16,18	16,12
	„Be jungčių“	15,26	11,49	5,35	6,35

lentelė 2 LTrSc upward ramping needs detalization

	Scenario	2028	2030	2033	2035
Number of MTUs showing an uncovered need, h	„Be jungčių ir P2X“	105	115	125	51
	„Be jungčių“	101	108	85	26
Interval between uncovered needs, h	„Be jungčių ir P2X“	111,8	155,3	103,2	215,6
	„Be jungčių“	117,9	161,8	147,2	467,3
Volume of uncovered needs in terms of energy, GWh	„Be jungčių ir P2X“	9,13	14,42	16,04	6,46
	„Be jungčių“	8,91	12,85	9,52	3,58

lentelė 3 NacSc base case analysis

Years	„NacSc bazinis“ evaluation			
	2028	2030	2033	2035
Number of MTUs showing an uncovered need, h	328,4	399,6	309,9	854,5
duration of uncovered needs, h	108	39	54	18
Volume of uncovered needs in terms of max power, MW	674,9	532,7	1082,8	539,5

lentelė 4 NacSc realistic case analysis

Metai	„NacSc realistinis“ evaluation			
	2028	2030	2033	2035

Number of MTUs showing an uncovered need, h	32,6	21,52	21,46	27,3
duration of uncovered needs, h	1111	1969	2170	1550
Volume of uncovered needs in terms of max power, MW	798,3	909	1135,5	1321,68

lentelė 5 LTrSc base case analysis

Metai	„LTrSc bazinis“ evaluation			
	2028	2030	2033	2035
Number of MTUs showing an uncovered need, h	0	0	0	0
duration of uncovered needs, h	0	0	0	0
Volume of uncovered needs in terms of max power, MW	0	0	0	0

lentelė 6 LTrSc realistic case analysis

Metai	„LTrSc realistinis“ evaluation			
	2028	2030	2033	2035
Number of MTUs showing an uncovered need, h	71,2	128,7	33,6	67,8
duration of uncovered needs, h	337	278	840	440
Volume of uncovered needs in terms of max power, MW	526	558.5	720,4	658,8